Supplementary Information
On The Conservation Laws and Pathway Realizability of Biochemical Reaction Networks

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1 Software architecture and analysis interfaces

Fig. S1 shows the input adapters and analysis interfaces of the chemical reaction network (CRN) submodule of `SynKit`. The shared reaction-system representation preserves reactant and product multiplicities, so derived stoichiometric matrices, Petri nets, and hypergraphs refer to the same reaction records and species identifiers. These representations connect model import and rule expansion to structural analysis and finite-inventory pathway verification.

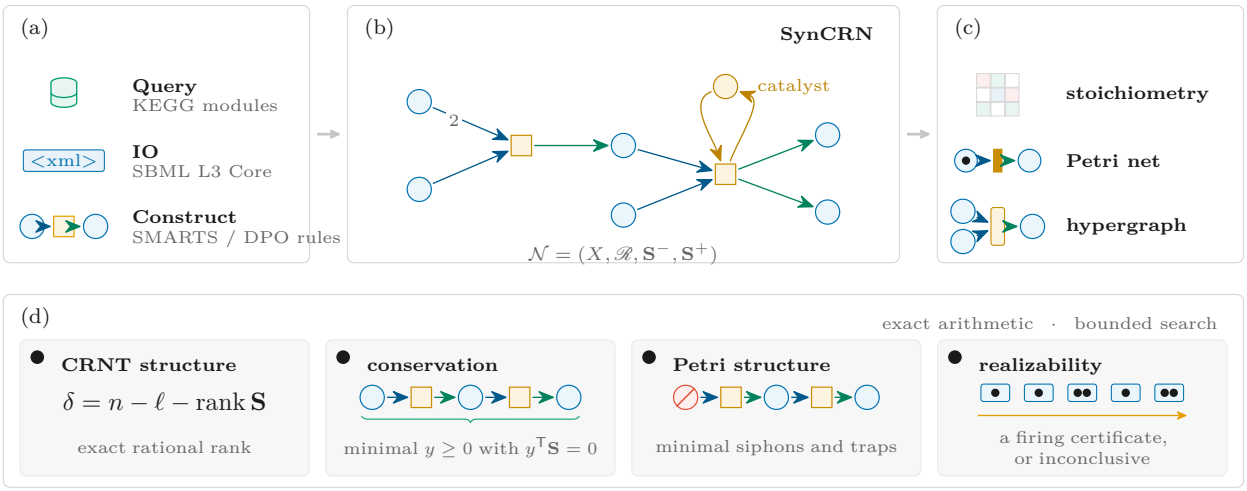


Fig. S1: Software architecture and data flow in the CRN submodule of `SynKit`. **(a)** Input adapters for `KEGG` pathways, `SBML` Core, and reaction rules. **(b)** Shared core records preserving reactant and product multiplicities. **(c)** Derived stoichiometric, Petri-net, and hypergraph representations. **(d)** Structural analyses: exact rational CRNT invariants, minimal conserved pools, Petri-net siphons/traps, and certified discrete pathway realizability (state/depth limits return `inconclusive`). Icons are schematic.

1.1 Python interface example

Listing S1 demonstrates the Python interface to the C++ backend using the glycolysis network. ATP/ADP/AMP, NAD(P)(H), water, phosphate, diphosphate, and protons are omitted in this example. The

resulting certificate therefore concerns a reduced model. Section 5 specifies the requested flux and the participant and currency controls.

```
from synkit.CRN import (build_native, native_backend, crnt_summary,
    conserved_moieties, find_siphons, siphon_persistence_details,
    PathwayRealizability, crn_to_sbml)
from Experiment.CRN.studies import build_kegg_crn, GLYCOLYSIS_FLUX

library = build_native("/tmp/synkit-crn", with_semiflows=True)
with native_backend(library):
    crn = build_kegg_crn("M00001", drop_currency=True)
    crn.n_species, crn.n_reactions                    # (14, 15)
    report = crnt_summary(crn)
    report.deficiency, report.is_weakly_reversible    # (1, False)
    len(conserved_moieties(crn)), len(find_siphons(crn)) # (3, 3)
    siphon_persistence_details(crn).persistence_ok    # False

    check = PathwayRealizability().load_syncrn_and_flow(
        crn, flow=GLYCOLYSIS_FLUX,
        initial_marking={"alpha-D-Glucose": 1},
        species="label", reaction="label")
    check.build_petri_net_from_flow()
    outcome = check.realizability_result()
    outcome.status.value, len(outcome.certificate) # ("realizable", 15)
    sbml_document = crn_to_sbml(crn)
```

Listing S1: Python analysis of reduced KEGG glycolysis (M00001): network construction, exact structural analysis, discrete pathway verification, and SBML export.

2 Reference-network validation and numerical audit

Table S1 lists the twelve reference networks, their literature sources, and the structural properties they validate. Nine are established models from biochemical and CRNT literature, while three are constructed diagnostic models that test linkage-class separation, non-unit stoichiometric coefficients, and the zero complex. These benchmark models serve as targeted correctness tests across distinct topological and algebraic configurations.

2.1 Algebraic consistency and deficiency validation

The validation combines expected analytical values, alternative algebraic formulations, and checks on returned basis vectors. Specifically, using the complex composition matrix $\mathbf{Y}$ and the directed complex-graph incidence matrix $\mathbf{I}_a$, the stoichiometric matrix satisfies $\mathbf{S} = \mathbf{Y}\mathbf{I}_a$. The deficiency reported from complexes (n), linkage classes (ℓ), and stoichiometric rank (s) must identically match the rank difference computed directly on the complex graph:

$$\delta \;=\; n - \ell - \operatorname{rank}(\mathbf{S}) \;=\; \operatorname{rank}(\mathbf{I}_a) - \operatorname{rank}(\mathbf{Y}\mathbf{I}_a). \tag{1}$$

Because the two sides compute intermediate quantities through independent algebraic pathways, this identity confirms internal consistency across graph partitioning, rank computation, and stoichiometric assembly.

Table S1: Reference reaction networks, sources, and targeted structural validation properties (invariants reported in Table 2).

Network model	**Source**	**Targeted algebraic / topological property**
Reversible isomerization	Feinberg [1], Ch. 3	Minimal weakly reversible system, Deficiency Zero Theorem (positive branch)
Irreversible isomerization	Feinberg [1], Ch. 3	Deficiency Zero Theorem (negative branch)
Isomerization cycle	Feinberg [1], Ch. 3	Weak reversibility without individual reversible steps
Three disjoint reversible pairs	Constructed test network	Linkage-class partitioning on disconnected graphs, where minimal semiflow supports must isolate disjoint pairs rather than their union
Disjoint pairs, non-unit stoichiometry	Constructed test network	As above, with $E \rightleftharpoons 2F$: exact conservation $2E + F$, which naive coefficient rounding can corrupt
Michaelis-Menten	Michaelis & Menten, Feinberg [1]	Archetypal enzyme mechanism with two conserved moiety pools, deficiency $\delta = 0$, non-weakly reversible
Reversible Michaelis-Menten	Feinberg [1]	Identical deficiency ($\delta = 0$), weak reversibility restored
Edelstein bistable switch	Edelstein [2]	Minimal reversible deficiency-one network admitting multiple steady states
Futile cycle (single-site)	Wang & Sontag [3]	Distributive enzymatic cycle with three independent conservation pools
Futile cycle (two-site)	Wang & Sontag [3]	Exact conservation relations in a multistationarity motif
Shinar-Feinberg ACR motif	Shinar & Feinberg [4]	Absolute concentration robustness, with a minimal siphon that contains no P-semiflow support
Open reactor ($0 \to A \to B \to 0$)	Constructed test network	Explicit zero-complex handling, where cycle closure changes weak reversibility

2.2 Numerical nullspaces and exact integer conservation laws

Conservation laws in reaction networks require non-negative integer vectors y satisfying $y^T\mathbf{S} = 0$. Standard numerical libraries (e.g., SVD or QR nullspace in `SciPy`) compute basis vectors in floating-point arithmetic, which inevitably leaves small numerical residuals. When floating-point basis vectors are scaled and rounded to nearest integers, the resulting vectors often fail the exact conservation condition.

We evaluated floating-point nullspace computation against `SynKit`'s exact integer arithmetic across the twelve reference networks (Table S2). While floating-point SVD leaves non-zero residuals on all closed networks, rounding fails to recover valid conservation vectors on three of the models (Michaelis–Menten, reversible Michaelis–Menten, and the one-site futile cycle). In contrast, `SynKit`'s exact rational solver yields integer vectors that satisfy $y^T\mathbf{S} = 0$ exactly in every case.

Table S2: Conservation residuals of numerical versus exact integer kernel vectors across the reference networks. SVD residual reports $\max|\mathbf{S}^T B|$ for floating-point basis B, Rounded invalid counts columns of round$(1000B)$ with non-zero integer residuals, and Exact residual evaluates `SynKit`'s integer basis. The open reactor has a zero-dimensional conservation space.

Network	SVD residual	Rounded invalid	Exact residual
Reversible isomerization	1.1e-16	0	0
Irreversible isomerization	0.0e+00	0	0
Isomerization cycle	2.2e-16	0	0
Disjoint pairs	1.1e-16	0	0
Non-unit disjoint pairs	1.1e-16	0	0
Michaelis-Menten	2.2e-16	1	0
Reversible Michaelis-Menten	1.7e-16	1	0
Edelstein	0.0e+00	0	0
One-site futile cycle	3.3e-16	3	0
Two-site futile cycle	5.3e-16	0	0
Shinar-Feinberg	1.1e-16	0	0
Open reactor	0.0e+00	0	0

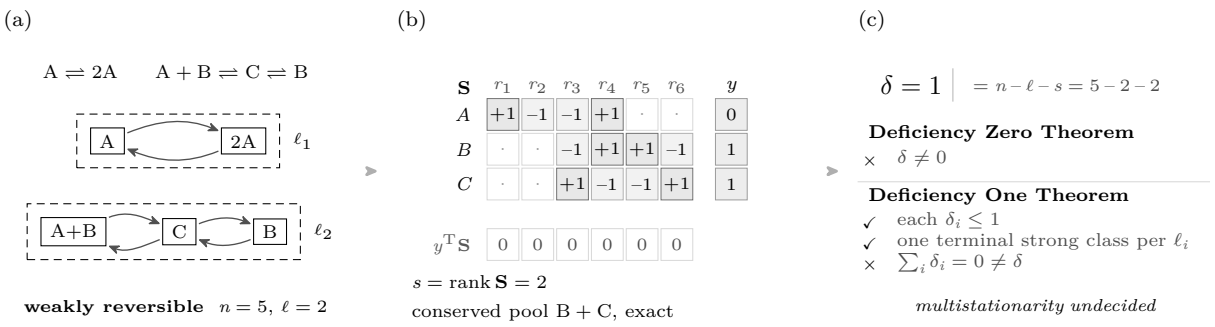


Fig. S2: Conservation structure and deficiency-theorem checks for the Edelstein network [2]. **(a)** Complex graph and its two linkage classes. **(b)** Stoichiometric matrix beside the primitive conservation vector y, whose product $y^{\mathsf{T}}\mathbf{S}$ vanishes over $\mathbb{Q}$ and certifies the conserved pool [B] + [C]. **(c)** Each classical deficiency theorem fails one hypothesis, leaving multistationarity undecided by deficiency alone.

3 Computational cost across network families

Computational scaling for minimal-siphon and conserved-pool enumeration was profiled across synthetic directed chains, reversible chains, and random sparse bimolecular networks. For minimal siphons, implication pruning avoids combinatorial subset explosion by selecting an initial species and propagating constraints: every reaction producing a chosen species must consume at least one member of the siphon.

In the random sparse family, conserved-pool enumeration completes at 395 species in 0.0514 s. Table S3 lists minimal-siphon counts and enumeration times for four seeded instances, showing that implication pruning scales tractably to hundreds of species. Fig. S3 summarizes the execution times across all structural analyses.

Table S3: Complete C++ minimal-siphon enumeration for the four seeded random-sparse instances. Each row is a single run; time includes the Python interface.

Species	Reactions	Minimal siphons	Time (s)
49	75	331	0.0554
99	150	272	0.0798
197	300	47	0.0177
395	600	87	0.158

4 BioModels: structural SBML interoperability

The eleven entries in Table S4 benchmark structural `SBML` exchange across a size-stratified selection from `BioModels`. Ten models contain reaction structures and pass export and re-import checks without loss of stoichiometric multiplicities, while `BIOMD0000000562` serves as a reaction-free control.

These benchmarks confirm that `SynKit` reliably extracts and preserves topological reaction networks and stoichiometric multiplicities across diverse curated models.

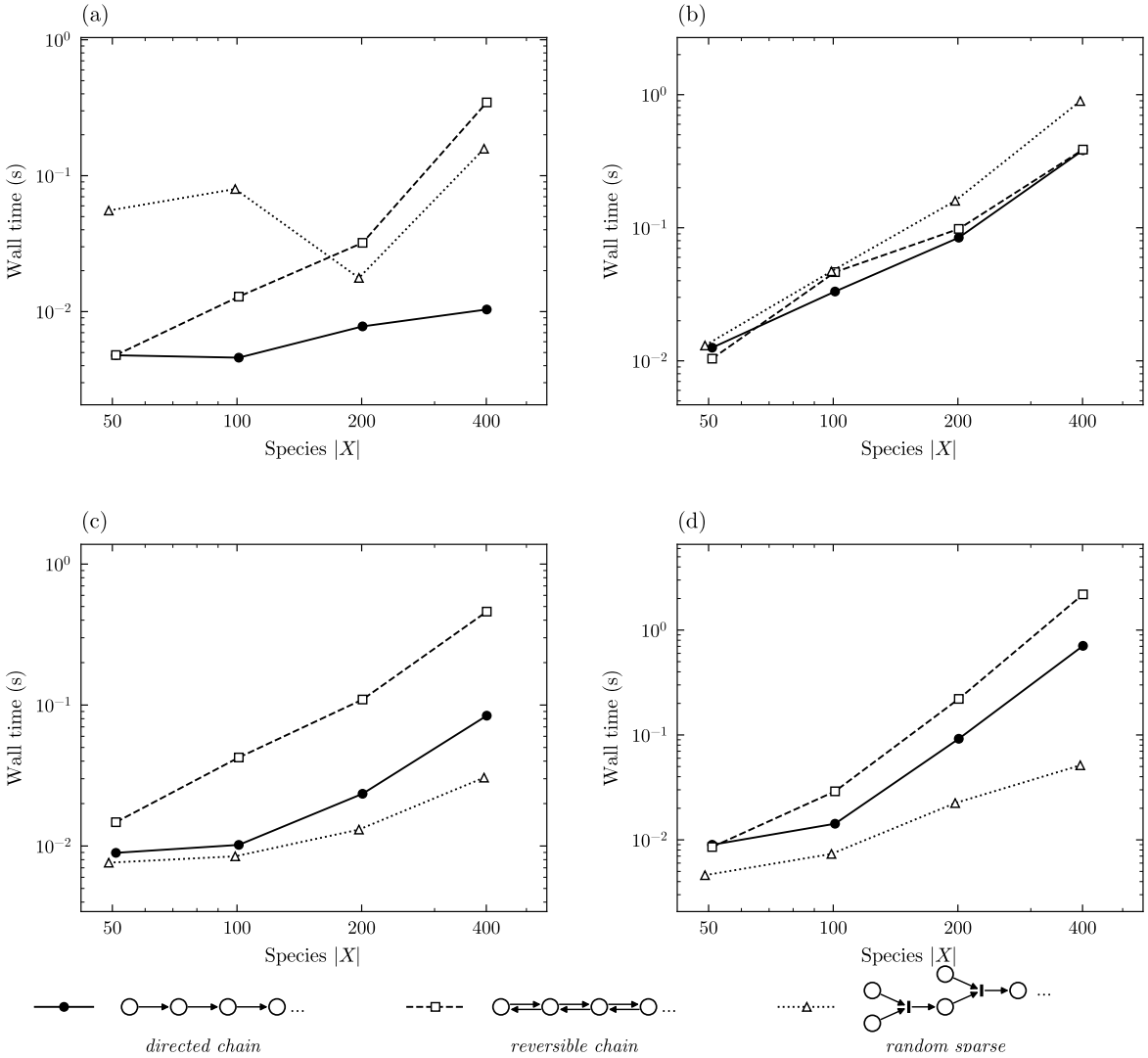


Fig. S3: C++ structural-analysis runtimes across directed-chain, reversible-chain, and random-sparse networks. **(a)** Minimal-siphon enumeration. **(b)** Exact CRNT analysis. **(c)** Graph canonicalization. **(d)** Exact conserved-pool enumeration. Points and connecting lines show completed single-run measurements. The key draws a small member of each family (circles: species, bars: bimolecular reactions) beside its line style. All 60 measurements complete.

Table S4: BioModels input inventory and structural results. m: species; r: directed reactions; s: exact rank; δ: deficiency; P: minimal conserved pools; S: minimal siphons; RT: supported structural round trip; Covered: every minimal siphon contains a conserved-pool support. Coverage is the sufficient persistence condition used in the main text, including the vacuous case of no siphons; failure does not prove extinction. Dashes mark analyses inapplicable to the reaction-free control.

Model	m	r	s	δ	P	S	RT	Covered
BIOMD0000000001	12	17	11	0	1	1	yes	no
BIOMD0000000010	8	10	5	0	3	3	yes	yes
BIOMD0000000051	18	48	18	8	0	0	yes	yes
BIOMD0000000064	26	17	17	0	11	8	yes	no
BIOMD0000000108	9	17	7	8	2	2	yes	yes
BIOMD0000000404	29	35	19	13	10	10	yes	no
BIOMD0000000637	12	35	6	0	6	3	yes	yes
BIOMD0000000019	100	125	81	41	19	20	yes	no
BIOMD0000000175	120	102	94	6	30	26	yes	no
BIOMD0000000255	504	827	486	317	17	75	yes	no
BIOMD0000000562	0	0	–	–	–	–	–	–

5 KEGG: module structure and glycolysis controls

The four cached modules in Table S5 are the inputs used in main-text Section 3.4: glycolysis (M00001), pyruvate oxidation (M00307), the citrate cycle (M00009), and the pentose phosphate pathway (M00004). The reported invariants refer to their reduced reaction equations with shared participants retained.

Table S5: Structural results for the four reduced KEGG modules. m: species; r: directed reactions; n: complexes; ℓ: linkage classes; s: rank; $\delta = n - \ell - s$: deficiency; P: minimal conserved pools; S: minimal siphons.

Module	m	r	n	ℓ	s	δ	P	S
M00001	14	15	16	4	11	1	3	3
M00307	13	8	14	7	6	1	8	6
M00009	22	12	23	11	12	0	12	9
M00004	12	12	14	4	10	0	2	4

5.1 Prescribed glycolytic flux and firing certificate

The main glycolysis model contains 14 species and 15 directed reactions, with currency metabolites omitted and shared polyphosphate and the ferredoxin bypass retained. The largest minimal conservation support comprises eleven carbohydrate intermediates, with coefficient two on hexoses and one on trioses.

Table S6 specifies the requested reaction multiset. Its ten reaction identifiers have a total multiplicity of 15; other reactions have zero requested multiplicity. The initial marking contains one alpha-D-Glucose token and zero tokens of every other species.

Table S6: Non-zero firing counts of the prescribed KEGG glycolysis flux. Identifiers match the cached M00001 reaction records.

Reaction	Count	Reaction	Count
R01786	1	R01061	2
R13199	1	R01512	2
R00756	1	R01518	2
R01068	1	R00658	2
R01015	1	R00200	2

Search explores 30 states and returns the following certificate, with reaction identifiers abbreviated to their numeric suffixes:

$$(01786, 13199, 00756, 01068, 01015, 01061, 01061, 01512,\\ 01512, 01518, 00658, 00200, 01518, 00658, 00200). \tag{2}$$

Independent verification confirms enabling at every firing, non-negative intermediate markings, and a final marking of two pyruvate tokens, while the weighted carbohydrate pool remains two throughout all 15 steps.

5.2 Shared-participant and currency controls

The shared-participant control requests one firing of `R02189` from one glucose token, comparing the retention of polyphosphate on both reaction sides with cancelling that common incidence. Species order, reaction order, and the net stoichiometric matrix remain identical. Table S7 demonstrates that retained polyphosphate must be present for enabling even though its net stoichiometric coefficient is zero.

Table S7: Shared-polyphosphate control for a single prescribed `R02189` firing. The glucose inventory is one in every row. n: complexes; ℓ: linkage classes; δ: deficiency.

Shared participant	PolyP tokens	n	ℓ	δ	Decision
Retained	0	16	4	1	blocked
Retained	1	16	4	1	fires
Cancelled	0	14	3	0	fires
Cancelled	1	14	3	0	fires

The currency control applies the 15-firing flux of Table S6 to a glucose-only initial marking. Restoring the omitted currency metabolites produces a 24-species model (Table S8). Without ATP supplied, hexokinase cannot fire and exhaustive search confirms that the pathway is unrealizable.

Table S8: Currency-boundary control for the prescribed full glycolytic flux from one glucose and zero tokens of all other species. The negative decision is exhaustive for this reaction multiset.

Currency metabolites	Species	Reactions	Decision	Certificate length
Removed	14	15	realizable	15
Retained	24	15	unrealizable	–

6 Formose: carbon conservation and precursor requirements

Four rules for aldol addition, retro-aldol cleavage, and reversible keto-enol tautomerization iteratively expand formaldehyde and glycolaldehyde seeds. The fourth iteration contains 37 species and 44 reactions, with 54 complexes, 17 linkage classes, rank 36, and deficiency one.

6.1 Network growth and carbon conservation

Table S9 reports the recorded species and reaction counts across iterations. The fourth iteration includes 31 species containing at least four carbons and reaches ten carbons.

Table S9: Network size and cumulative generation time from the seeds through each expansion depth in the original single-run formose study. These timings include first-call overhead; Table S10 provides the repeated comparison after warm-up.

Iteration	Species	Reactions	Cumulative time (s)
1	3	1	0.029
2	5	4	0.003
3	9	10	0.004
4	37	44	0.010

The stoichiometric matrix $\mathbf{S} \in \mathbb{Z}^{37\times44}$ has a one-dimensional left nullspace whose positive primitive conservation vector agrees exactly with the molecular carbon count for every species, confirming that the generated reaction rules strictly conserve carbon.

6.2 Independent generation checks and expansion runtimes

The `SynKit` timings in the cross-tool comparison use the native CRN expander with Lewis hydrogen counts. Table S11 separately compares this implementation with native explicit-hydrogen expansion and the Python/RDKit construction path, so that backend and hydrogen-model differences remain visible.

The same four element-and-bond rules are represented as mapped reaction `SMARTS` in `SynKit` and explicit-hydrogen DPO rules in `MØD` 1.0.0.75 and `MolFoundry` 0.1.11. The translation preserves mapped atoms and the bonds removed, retained, and added. Both seeds are initially active. Each parallel expansion round requires at least one frontier reactant and excludes mixtures containing two copies of the same species, matching the published `SynKit` configuration. This mixture policy is a generation setting, not a chemical prohibition. No molecular-size cutoff is imposed. The `SynKit` task and mixture caps are raised to 2^{60}, and completed runs are checked for truncation.

Each tool and depth runs in an isolated process pinned to the same logical CPU of an Intel Core Ultra 7 155H, with one warm-up followed by five timed expansions from fresh seeds. Timings include rule and seed parsing, expansion, and construction of each tool's network representation. They exclude imports, engine startup, the fixed `SMARTS`-to-GML translation, cross-tool identity checks, and structural analyses. `SynKit` uses its native Lewis-hydrogen expander (RDKit C++ 2026.03.1); the other tools use explicit-hydrogen graph rewriting. The measurements therefore compare complete generation interfaces, including their representation and language-interface costs. Python versions, dependency lists, native build fingerprints, all samples, and exported network snapshots are saved with the benchmark evidence.

For each completed run, exported molecular graphs are converted to canonical non-stereochemical SMILES with explicit hydrogens removed. Reactions are compared as directed pairs of reactant and product multisets, preserving multiplicities. The identity check ignores atom maps, rule labels, and alternative atom-level derivations of the same stoichiometric reaction. All three tools return identical species and directed stoichiometric reaction sets at every depth from one through six, including 302 species and 371 reactions at depth five and 10 572 species and 11 239 reactions at depth six. Carbon balance is independently checked for every exported reaction. Median times and the first and third quartiles are reported in Table S10; Fig. S4 shows cross-tool runtimes and the implementation controls.

Depth seven does not yield a complete network in the cross-tool comparison under these limits. The `SynKit` native Lewis worker times out at 180 s during its first expansion; `MØD` raises `MemoryError` after approximately 176 s; and `MolFoundry`'s JavaScript runtime reports an out-of-memory failure and the worker exits with signal 11 after approximately 49 s. Counts and generation-time estimates are therefore withheld

for that depth. These resource-limited attempts do not establish an intrinsic maximum expansion depth for any tool.

Table S10: Cumulative formose generation times in seconds: median [first quartile, third quartile] over five runs after one warm-up. Counts describe the completed networks; identity is tested using the complete species and stoichiometric reaction sets. Each worker has a 180 s wall-clock budget covering startup, warm-up, five expansions, and identity audits, and an 8 GiB per-process address-space limit. A timeout or memory failure is an incomplete measurement, not evidence of chemical closure or target unreachability.

Depth	Species	Reactions	`SynKit` native (Lewis H)	`MØD`	`MolFoundry`
1	3	1	0.0004862 [0.0004809, 0.0004924]	0.0003832 [0.000381, 0.0003981]	0.0121 [0.0112, 0.01235]
2	5	4	0.0008686 [0.0008619, 0.0008993]	0.0005635 [0.0005559, 0.0005639]	0.02453 [0.01532, 0.03018]
3	9	10	0.001607 [0.001603, 0.001689]	0.000969 [0.0009682, 0.0009777]	0.02933 [0.01364, 0.03301]
4	37	44	0.006683 [0.006673, 0.006729]	0.002895 [0.002865, 0.002898]	0.03722 [0.03364, 0.04229]
5	302	371	0.07579 [0.07527, 0.07657]	0.02843 [0.02816, 0.02858]	0.1745 [0.1684, 0.1745]
6	10572	11239	4.293 [4.259, 4.3]	1.575 [1.566, 1.605]	11.64 [11.5, 11.7]
7	–	–	timeout	memory limit	OOM crash

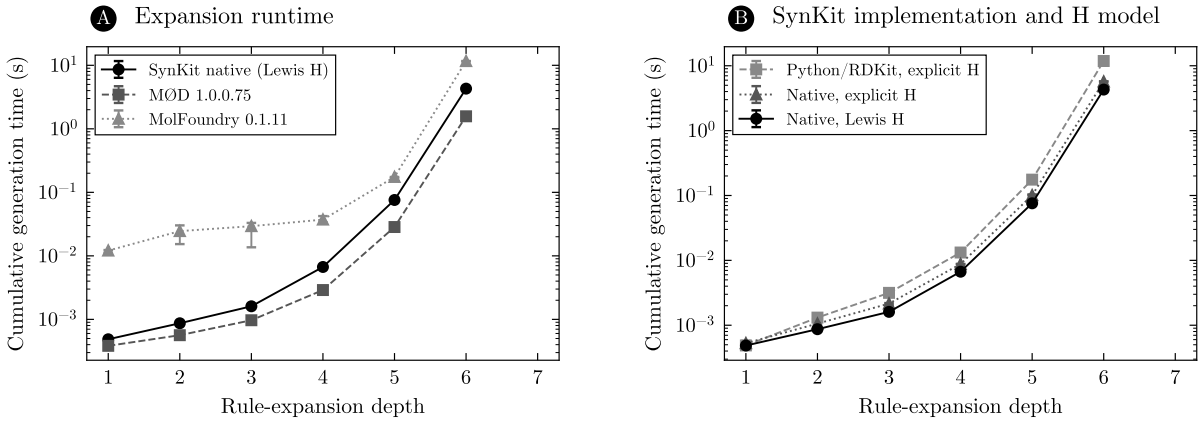


Fig. S4: Formose expansion benchmark. **(a)** Median cumulative generation time with first-to-third-quartile error bars over five runs after warm-up. **(b)** Controlled comparison of `SynKit`'s Python/RDKit construction path, native explicit-hydrogen expansion, and native Lewis-hydrogen expansion. Lines connect measured depths; incomplete depth-seven runs have no timing point.

Table S11: Implementation and hydrogen-representation controls in `SynKit` 1.6.5. Cumulative generation time in seconds is reported as median [first quartile, third quartile] across five runs after warm-up. The Python control uses `CRNExpand(rule_backend="rdkit")`; both native controls use `rule_backend="native"`, changing only `hydrogen_model`. The Python/RDKit control uses RDKit 2025.09.3, whereas both native controls use RDKit C++ 2026.03.1, so the Python–native comparison also includes an RDKit-version difference. The native Lewis column repeats the original comparison measurements. All completed controls reproduce the same complete molecular species and stoichiometric reaction sets. The worker limits match Table S10.

Depth	Python/RDKit, explicit H	Native, explicit H	Native, Lewis H
1	0.000488 [0.0004786, 0.0005746]	0.0005279 [0.0005199, 0.0005556]	0.0004862 [0.0004809, 0.0004924]
2	0.001311 [0.001284, 0.001317]	0.001058 [0.001026, 0.001156]	0.0008686 [0.0008619, 0.0008993]
3	0.003154 [0.003039, 0.003355]	0.002156 [0.002067, 0.002196]	0.001607 [0.001603, 0.001689]
4	0.01321 [0.01246, 0.01363]	0.008774 [0.008767, 0.008798]	0.006683 [0.006673, 0.006729]
5	0.1756 [0.173, 0.1761]	0.1009 [0.1006, 0.1039]	0.07579 [0.07527, 0.07657]
6	11.84 [11.75, 11.85]	5.7 [5.692, 5.747]	4.293 [4.259, 4.3]
7	memory limit	timeout	timeout

At depth six, median generation times are 11.8 s for Python/RDKit, 5.7 s for native explicit hydrogen, and 4.29 s for native Lewis hydrogen. The native Lewis path is 1.33 times faster than the native explicit-hydrogen path and 2.76 times faster than the Python/RDKit path. A separate instrumented native Lewis control has a median total time of 4.27 s, with 4.09 s in the native request, including transport and

decoding, and 0.163 s in Python-side network construction. The native-request measurement combines reaction matching, product sanitization, molecular canonicalization, and protocol costs.

6.3 Route definitions and finite precursor inventories

The targets are an aldotetrose (T_4, `O=CC(O)C(O)CO`) and an aldohexose (T_6, `O=CC(O)C(O)C(O)C(O)CO`). Under qualitative set reachability, both are reachable from the two seed species. Table S12 distinguishes qualitative depth from the reaction multiplicities required for discrete firing, and Fig. S5 shows the reaction motifs.

Table S12: Qualitative depths and prescribed discrete reaction multisets for the selected formose targets. Set reachability reuses a species once generated; discrete firing requires a separate token for each consumed molecule.

Target	Qualitative depth	Discrete firings	Prescribed multiset
T_4	2	2	$r_1 + r_3$
T_6	3	4	$2r_1 + r_3 + r_7$

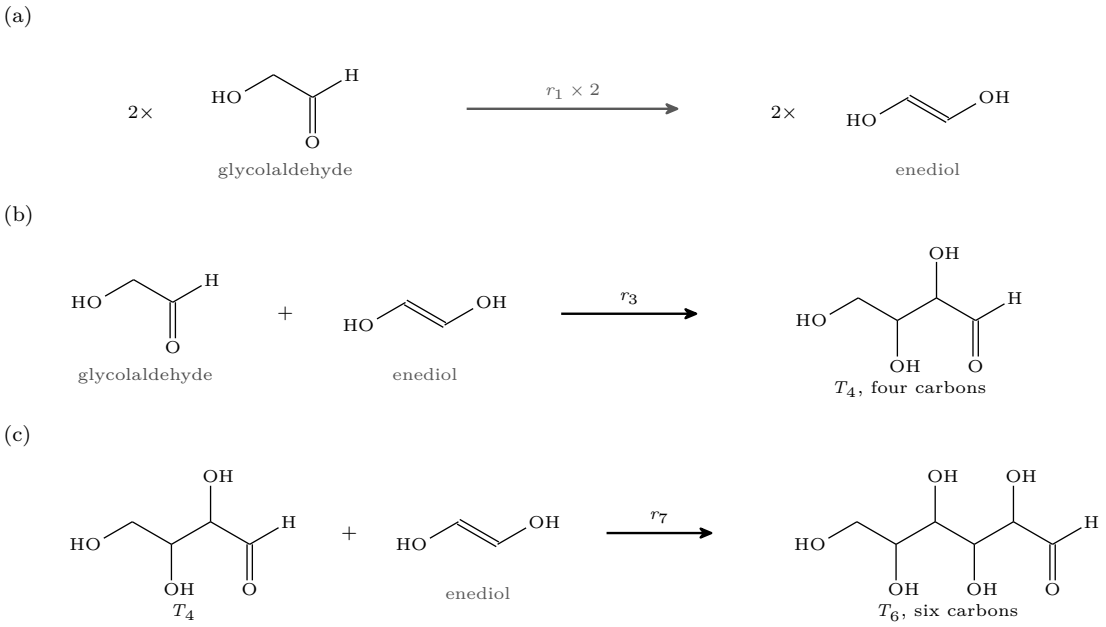


Fig. S5: Stoichiometry and precursor requirements of the selected formose routes. **(a)** Tautomerization (r_1) of glycolaldehyde to enediol. **(b)** Aldol coupling (r_3) to the aldotetrose (T_4). **(c)** Second aldol coupling (r_7) to the aldohexose (T_6). Firing counts are per route. Structures are constitutional; no stereochemistry or atom mapping is assigned.

Table S13 evaluates five initial precursor inventories. Formaldehyde is C_1 and glycolaldehyde is C_2. Every positive sequence is confirmed by an executable firing sequence, while negative decisions are established by exhaustive search over the prescribed multiset. For an independent check, the same reaction multisets and initial markings are supplied directly to `MolFoundry`'s `FlowSolution`, without optimizing a new hyperflow. Its `is_realizable()` method and SMT certificate enumeration agree with all ten `SynKit` decisions. All three positive certificates are independently replayed, checking enabling before every firing, the requested

reaction multiplicities, and target production. This comparison concerns the prescribed routes, rather than unrestricted target reachability.

Table S13: Realizability of the prescribed tetrose and hexose routes across five precursor inventories. Counts denote molecules; total carbon counts carbon tokens. Yes indicates a verified certificate and No an exhaustive negative decision for the selected multiset.

Pool	C_1 count	C_2 count	Total carbon	T_4	T_6
M_1	1	1	3	no	no
M_2	2	1	4	no	no
M_3	4	1	6	no	no
M_4	2	2	6	yes	no
M_5	0	3	6	yes	yes

The tetrose route converts glycolaldehyde to enediol (r_1), then couples that enediol with a second glycolaldehyde (r_3). It requires two glycolaldehyde molecules and leaves formaldehyde untouched; pools M_4 and M_5 support it, whereas M_3 (which contains only one glycolaldehyde) cannot. The hexose route requires two enediol tokens, consuming three glycolaldehyde molecules in total ($2r_1 + r_3 + r_7$), which only pool M_5 satisfies. The three six-carbon inventories (M_3, M_4, and M_5) thus produce different realizability outcomes despite having equal total carbon budgets.

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